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Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

School of Computer Science Engineering & Information Systems (SCORE)

WINTER SEMESTER 2025-2026

Name: M. Divyasri

Registration Number: 23MIS0174

Course Title: Agile Development Process and Devops Lab

Course Code: ISWE406P

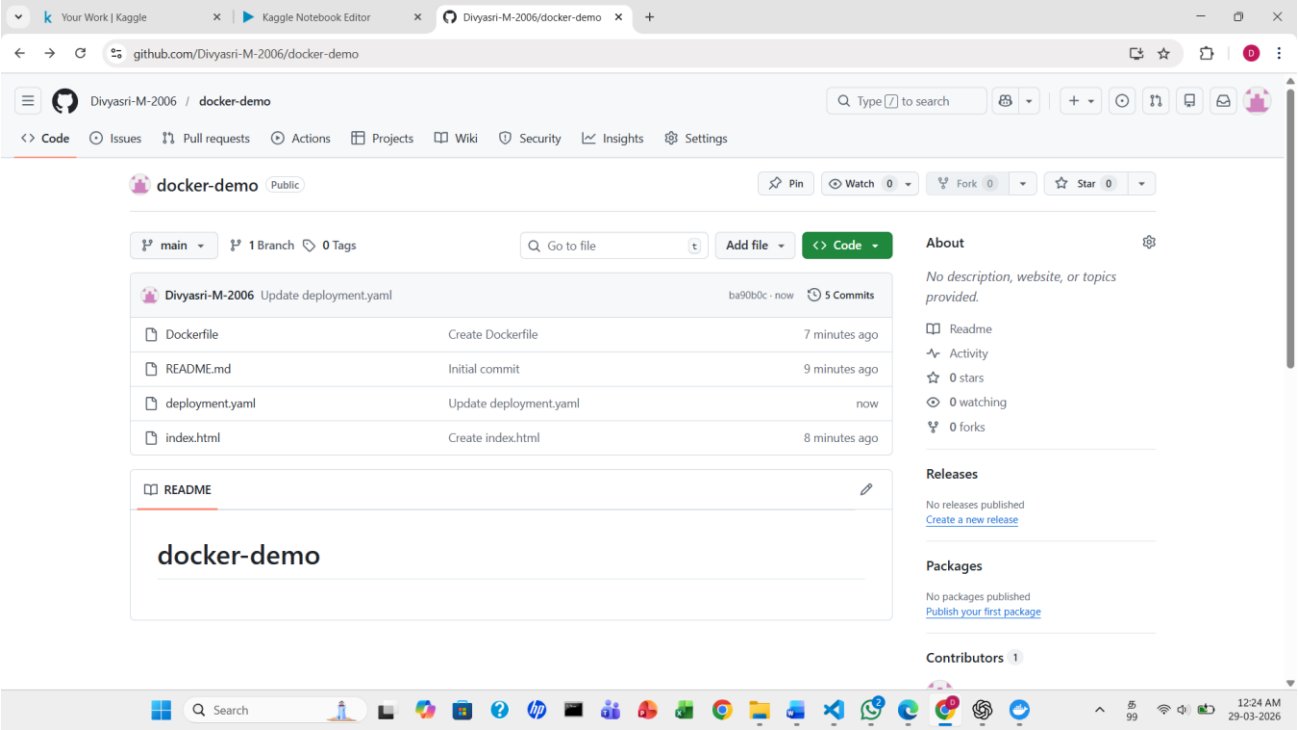
Faculty: Dr. Sweta Bhattacharya

Slot: L3+L4

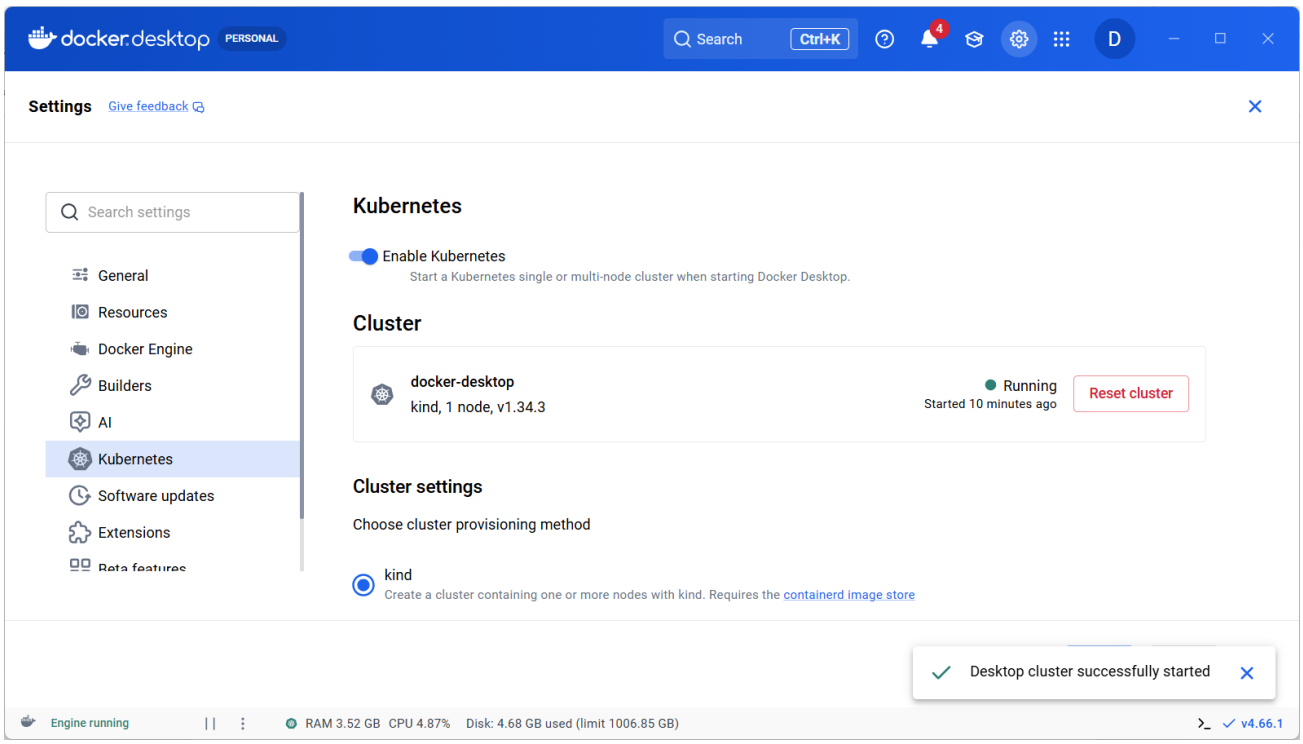
Assignment: Create an HTML file for your profile and complete the following:

1. Create a repository in GIT and create the following files:

- **index.html**
- **Dockerfile**
- **Deployment.yaml**



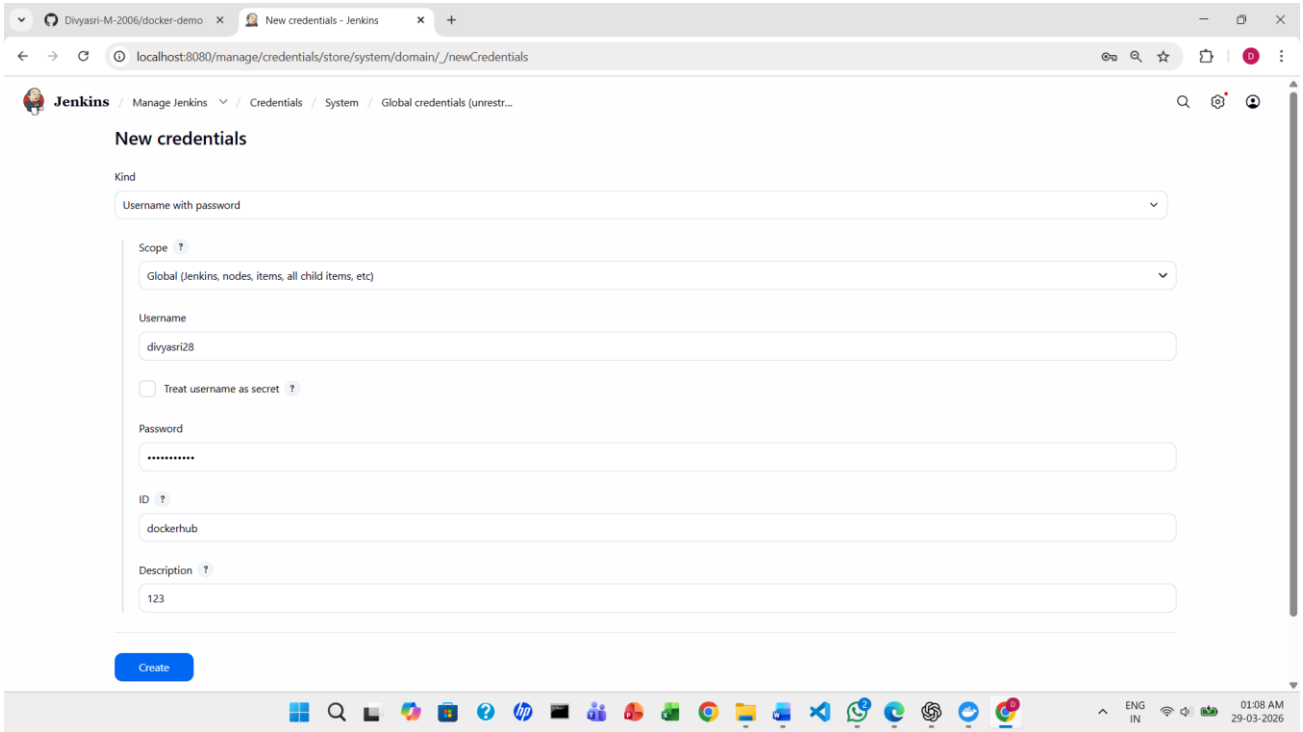
2. Enable Kubernetes
   - **Open the Docker Dashboard**: Launch the Docker Desktop application on your system.
   - **Navigate to Settings**: Click the gear icon (Settings or Preferences) in the top-right or sidebar of the Docker Dashboard.
   - **Select the Kubernetes Tab**: In the Settings menu, find and select the **Kubernetes** tab.
   - **Enable Kubernetes**: Check the box labeled **Enable Kubernetes**.
   - **Apply & Restart**: Click the **Apply & Restart** button at the bottom of the settings window. Docker Desktop will then download the necessary Kubernetes components and start a single-node cluster. This may take a few minutes the first time



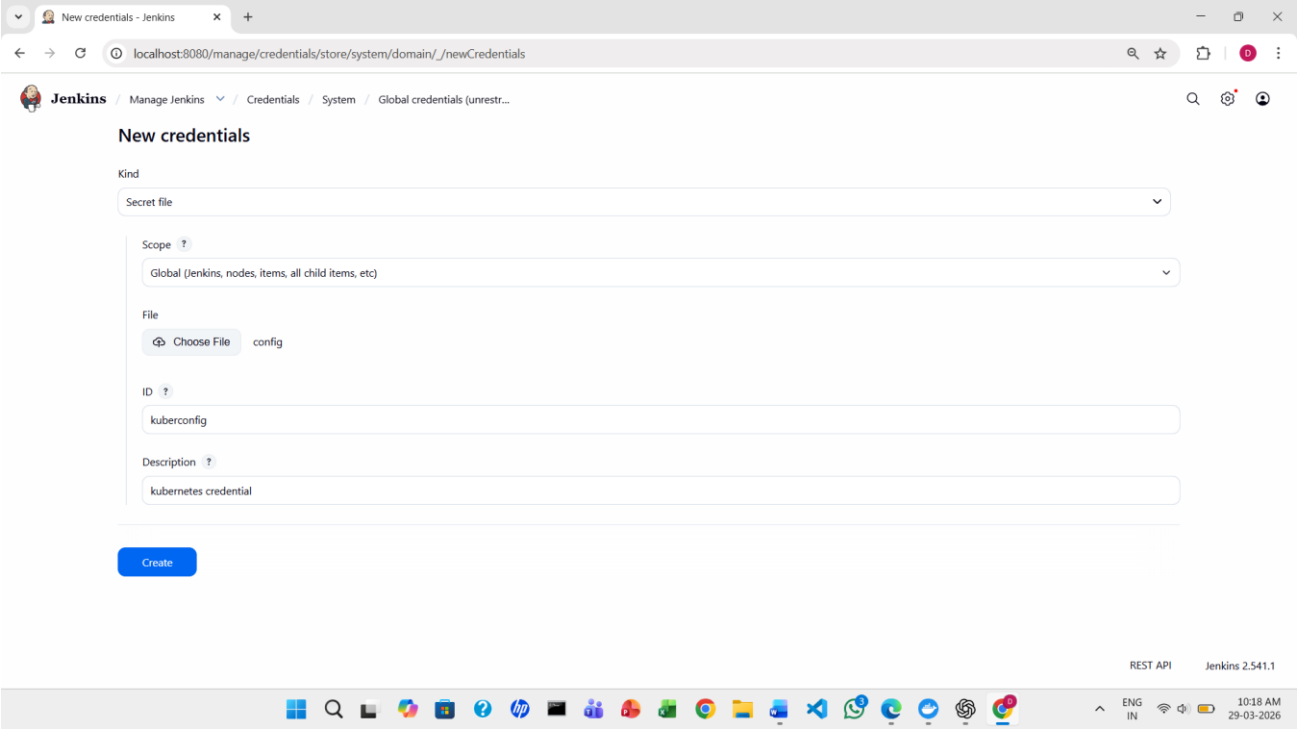
3. Open Jenkins

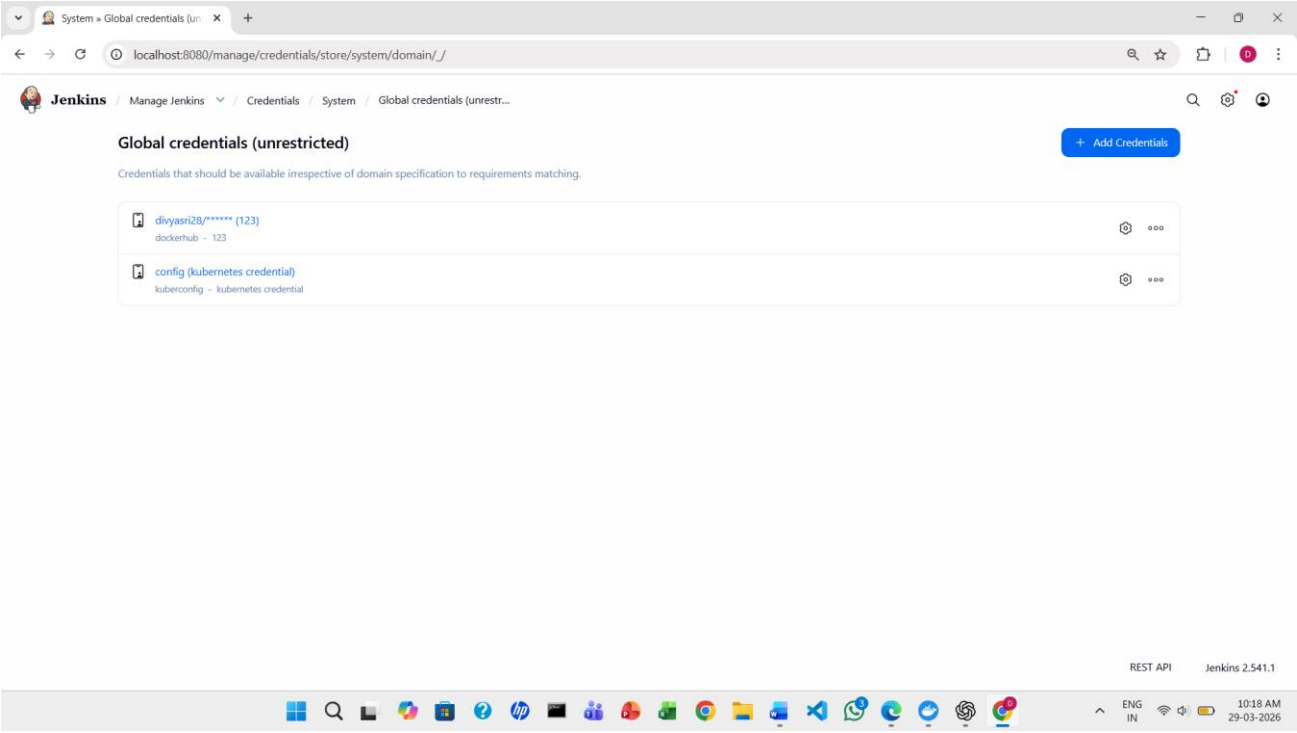
In Jenkins Setting --> Credentials

New Credentials – User name: <should use the Docker Desktop username>, password <same as Docker Desktop password>, ID:<dockerhub>, Description <any>



New Credentials – Secret – Browse C: User->Administrator-->.kube

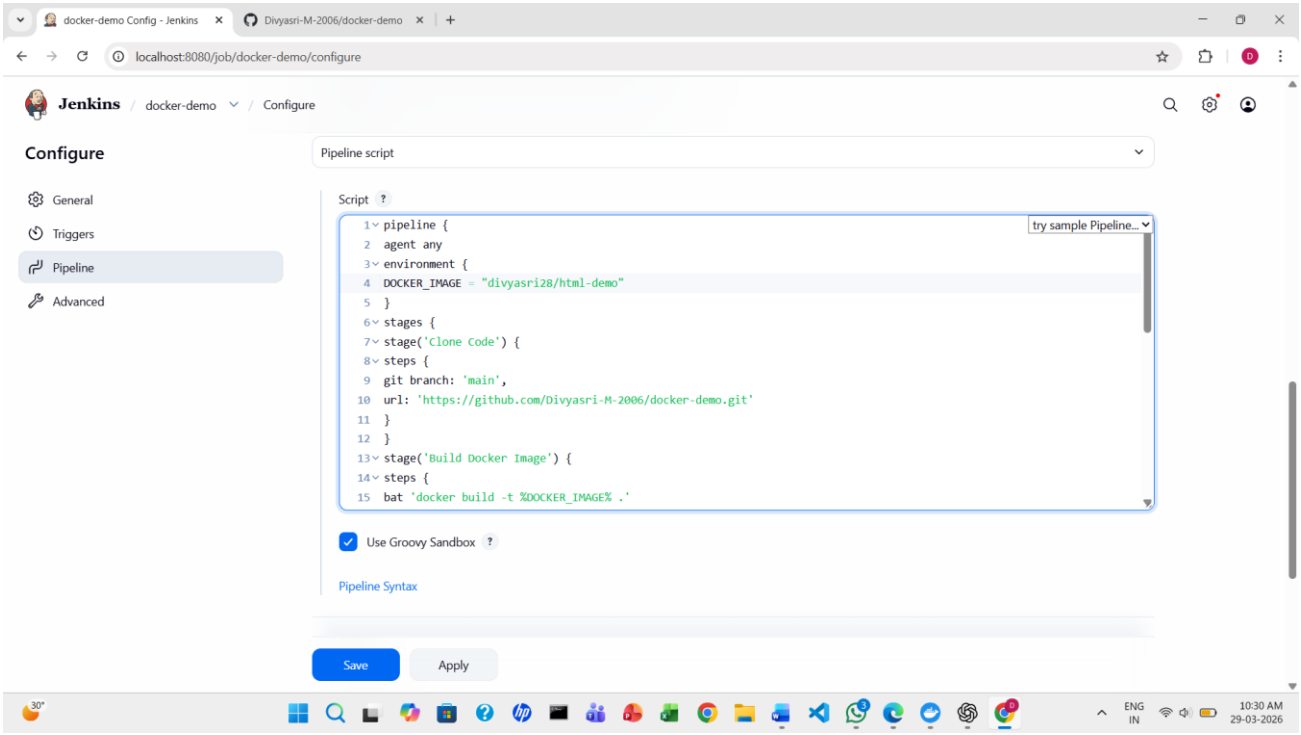


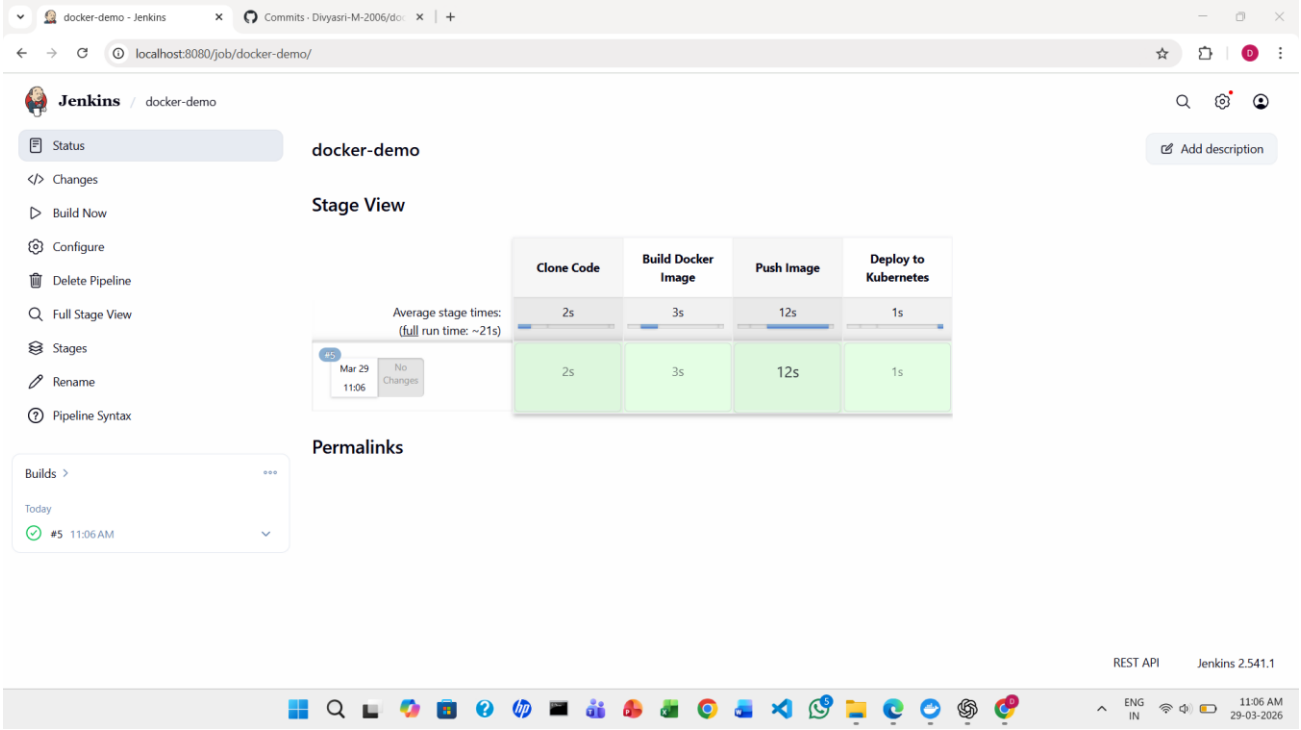


Create New Item -> Pipeline

Paste the code in pipeline script

- **Update URL of GIT repository**
- **Make sure credentialsId: “dockerhub” same as mentioned in credentials**
- **Make sure that the ID mentioned in credentials “kuberconfig” is same as in script - Run the Build Pipeline:**





Console - #5 - Jenkins | Commits · Divyasri-M-2006/doc

localhost:8080/job/docker-demo/5/console

Jenkins / docker-demo / #5

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#5'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

Console

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```
Started by user Divyasri_M
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\ProgramData\Jenkins\.jenkins\workspace\docker-demo
[Pipeline] {
[Pipeline] withEnv
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Clone Code)
[Pipeline] git
The recommended git tool is: NONE
No credentials specified
 > C:\Program Files\Git\cmd\git.exe rev-parse --resolve-git-dir C:\ProgramData\Jenkins\.jenkins\workspace\docker-demo\.git # timeout=10
Fetching changes from the remote Git repository
 > C:\Program Files\Git\cmd\git.exe config remote.origin.url https://github.com/Divyasri-M-2006/docker-demo.git # timeout=10
Fetching upstream changes from https://github.com/Divyasri-M-2006/docker-demo.git
 > C:\Program Files\Git\cmd\git.exe --version # timeout=10
 > git --version # 'git version 2.52.0.windows.1'
 > C:\Program Files\Git\cmd\git.exe fetch --tags --force --progress -- https://github.com/Divyasri-M-2006/docker-demo.git
+refs/heads/*:refs/remotes/origin/* # timeout=10
 > C:\Program Files\Git\cmd\git.exe rev-parse "refs/remotes/origin/main^{commit}" # timeout=10
Checking out Revision 7fffb99c4dc063babe040e3bbc575f796dec092c (refs/remotes/origin/main)
 > C:\Program Files\Git\cmd\git.exe config core.sparsecheckout # timeout=10
 > C:\Program Files\Git\cmd\git.exe checkout -f 7fffb99c4dc063babe040e3bbc575f796dec092c # timeout=10
 > C:\Program Files\Git\cmd\git.exe branch -a -v --no-abbrev # timeout=10
```

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Console - #5 - Jenkins | Commits · Divyasri-M-2006/doc

localhost:8080/job/docker-demo/5/console

Jenkins / docker-demo / #5

```
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy to Kubernetes)
[Pipeline] withCredentials
Masking supported pattern matches of %KUBECONFIG%
[Pipeline] {
[Pipeline] bat

C:\ProgramData\Jenkins\.jenkins\workspace\docker-demo>set KUBECONFIG=****

C:\ProgramData\Jenkins\.jenkins\workspace\docker-demo>kubectl apply -f deployment.yaml --validate=false
deployment.apps/html-demo created
service/html-demo-service created
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.541.1

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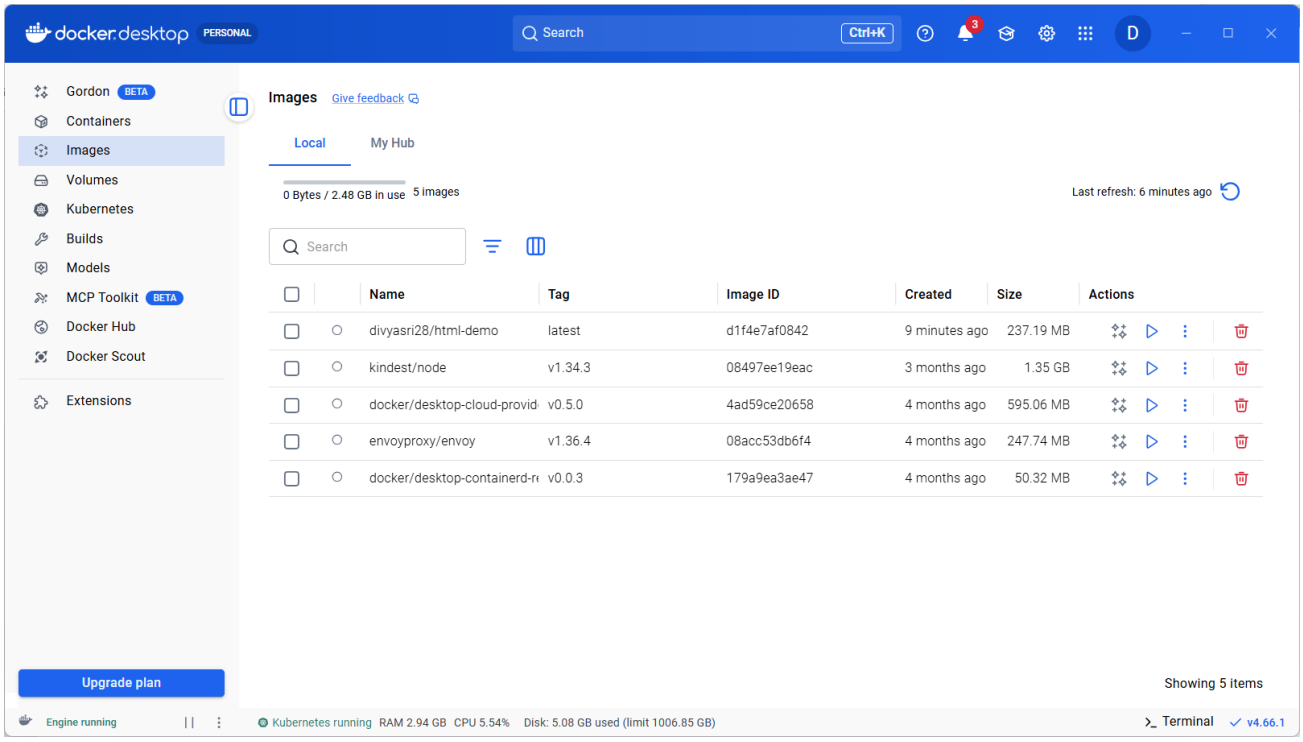
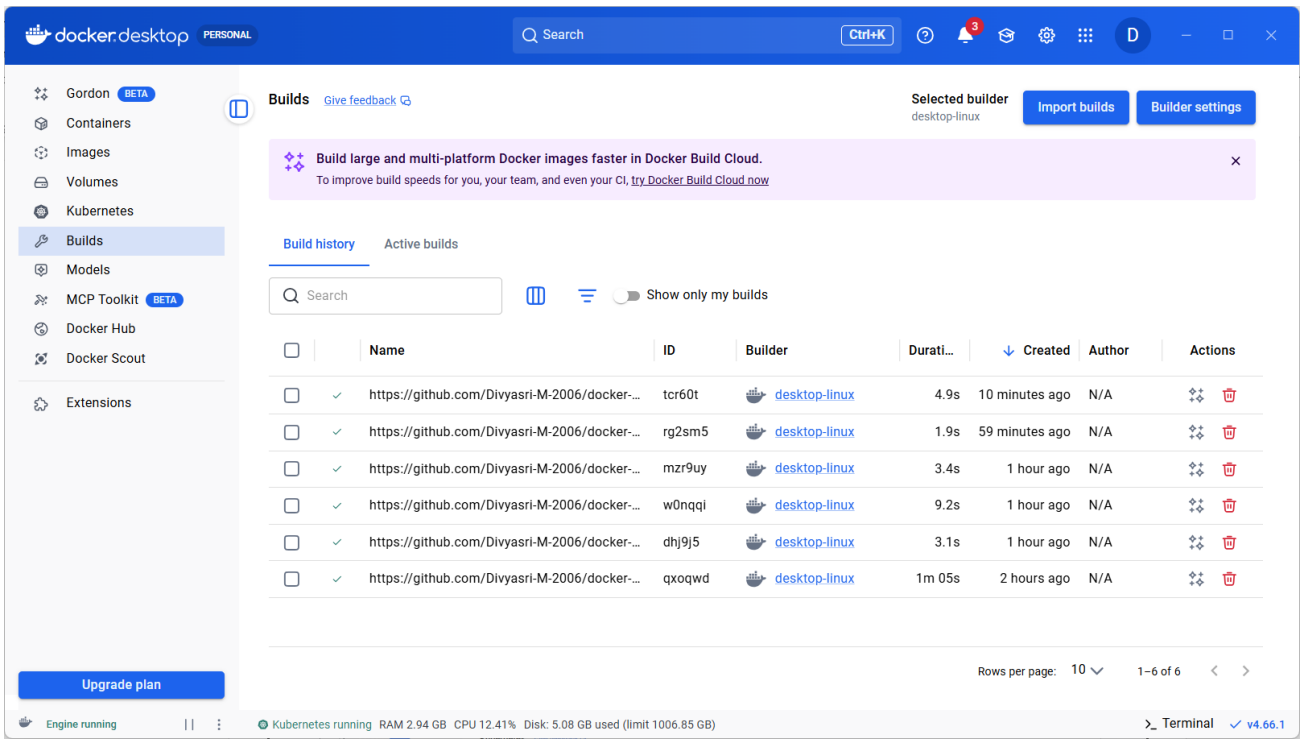
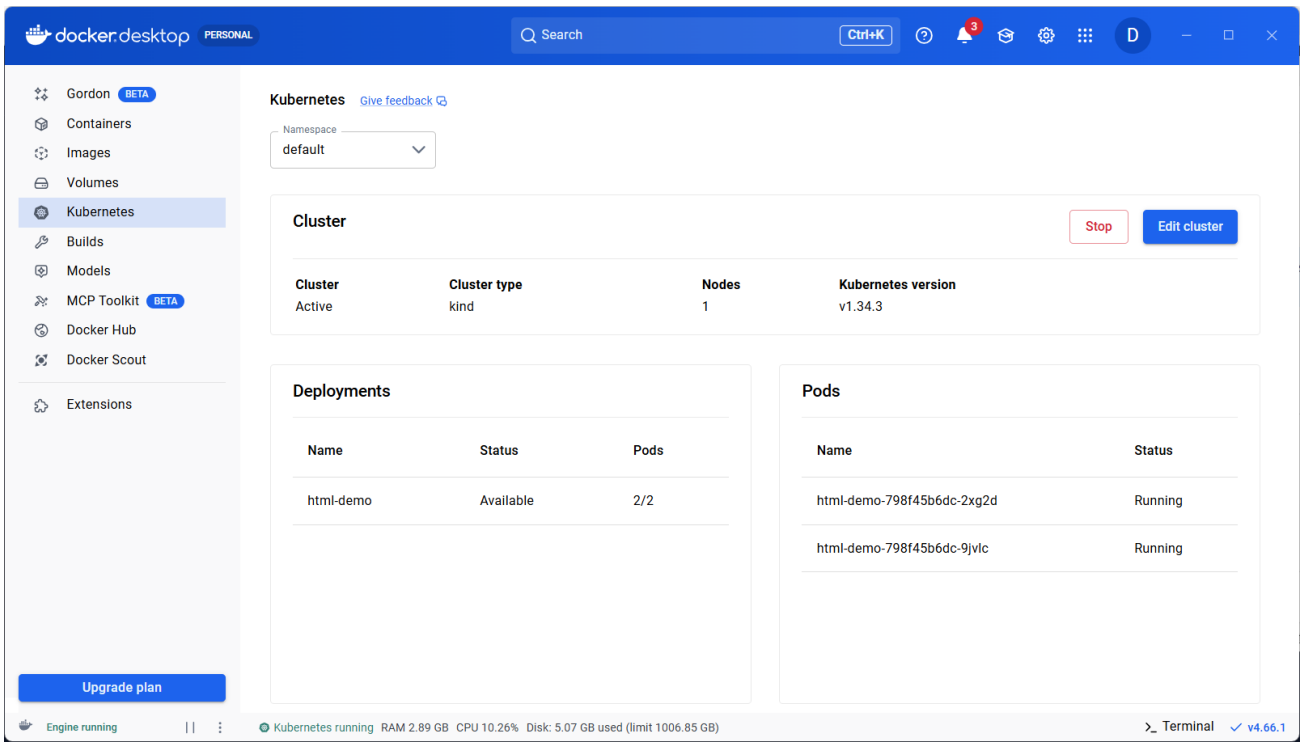
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\nivet> kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
html-demo-798f45b6dc-2xg2d    1/1     Running   0          5m13s
html-demo-798f45b6dc-9jvlc    1/1     Running   0          5m13s
PS C:\Users\nivet> kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP   PORT(S)        AGE
html-demo-service   NodePort    10.96.97.25   <none>        80:30010/TCP   5m14s
kubernetes          ClusterIP   10.96.0.1     <none>        443/TCP        71m
PS C:\Users\nivet>
```

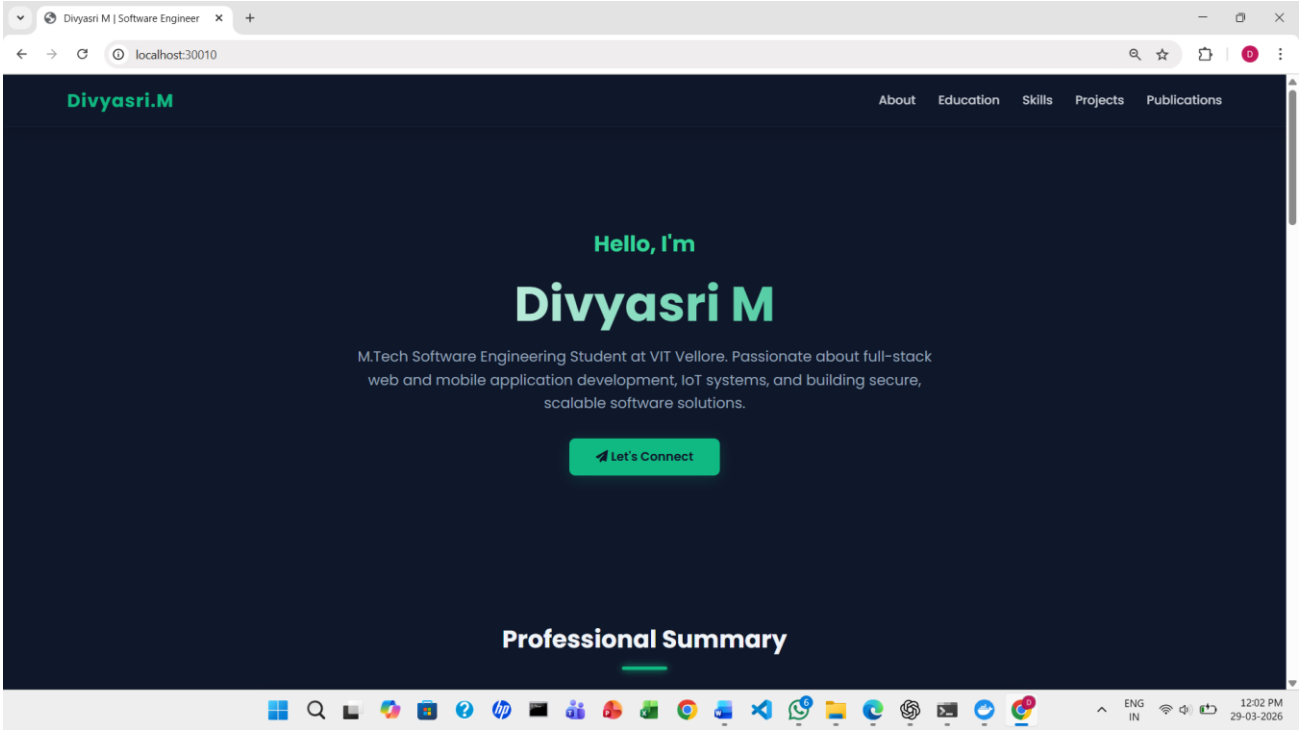
Dokerdesktop View

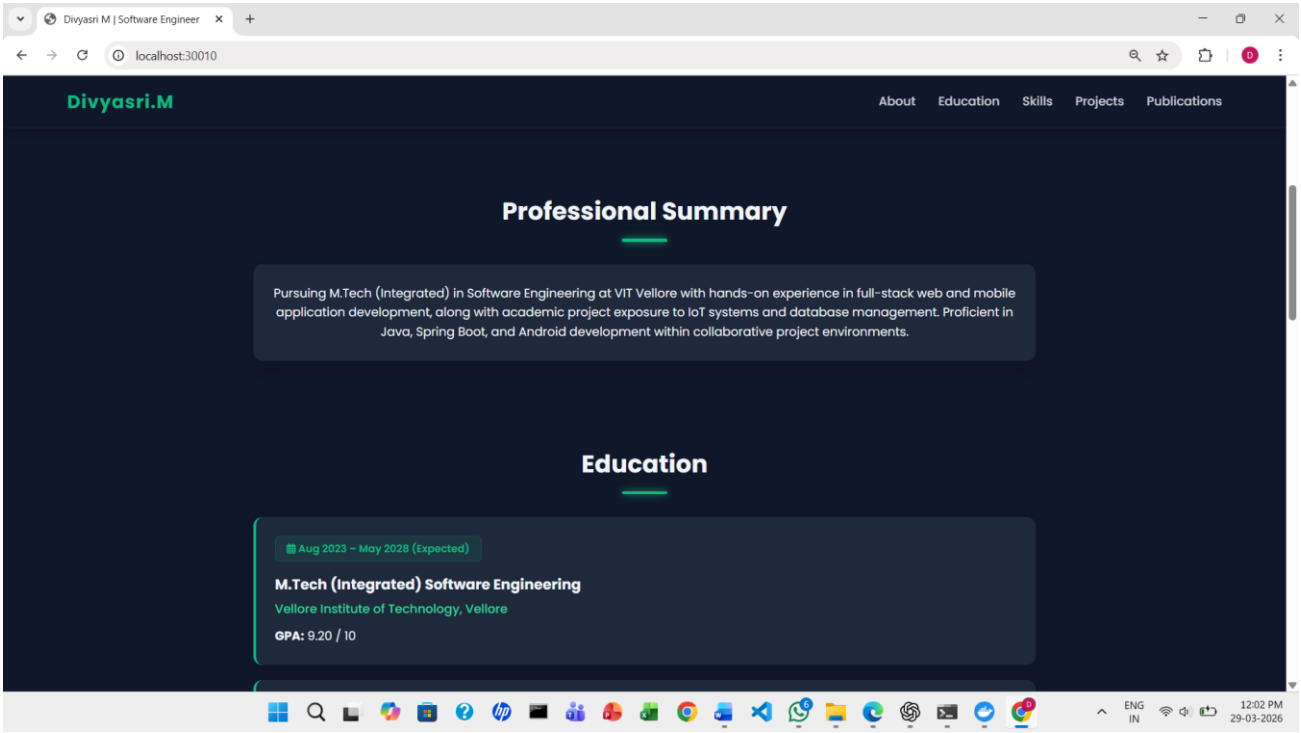


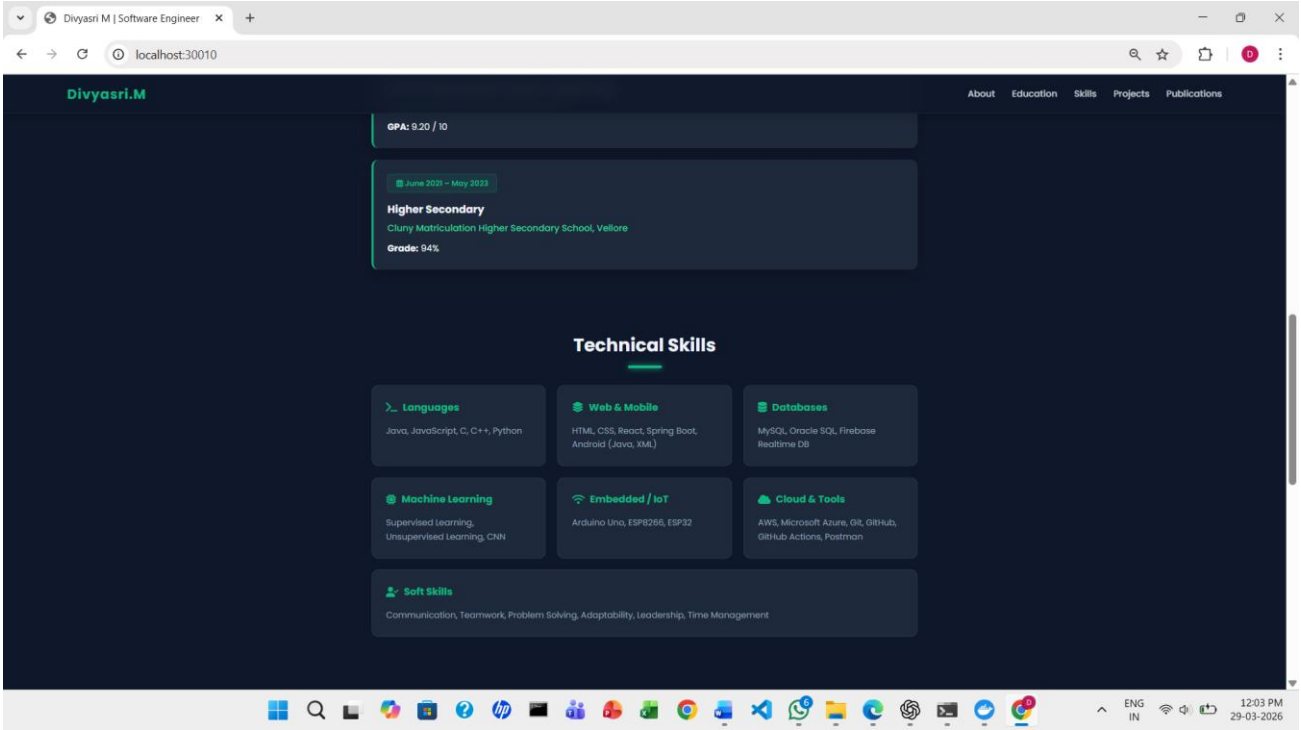

docker desktop PERSONAL
Search Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Builds Give feedback
Selected builder
desktop-linux
Import builds
Builder settings
Build large and multi-platform Docker images faster in Docker Build Cloud.
To improve build speeds for you, your team, and even your CI, try Docker Build Cloud now
Build history
Active builds
Search
Show only my builds
Name ID Builder Durati... Created Author Actions
https://github.com/Divyasri-M-2006/docker-... tcr60t desktop-linux 4.9s 10 minutes ago N/A
https://github.com/Divyasri-M-2006/docker-... rg2sm5 desktop-linux 1.9s 59 minutes ago N/A
https://github.com/Divyasri-M-2006/docker-... mzr9uy desktop-linux 3.4s 1 hour ago N/A
https://github.com/Divyasri-M-2006/docker-... w0nqqi desktop-linux 9.2s 1 hour ago N/A
https://github.com/Divyasri-M-2006/docker-... dhj9j5 desktop-linux 3.1s 1 hour ago N/A
https://github.com/Divyasri-M-2006/docker-... qxoqwd desktop-linux 1m 05s 2 hours ago N/A
Rows per page: 10
1–6 of 6
Engine running
Kubernetes running RAM 2.94 GB CPU 12.41% Disk: 5.08 GB used (limit 1006.85 GB)
Terminal
v4.66.1


docker desktop PERSONAL
Search Ctrl+K
Gordon BETA
Containers
Images
Volumes
Kubernetes
Builds
Models
MCP Toolkit BETA
Docker Hub
Docker Scout
Extensions
Upgrade plan
Kubernetes Give feedback
Namespace
default
Cluster
Stop
Edit cluster
Cluster Active
Cluster type kind
Nodes 1
Kubernetes version v1.34.3
Deployments
Name Status Pods
html-demo Available 2/2
Pods
Name Status
html-demo-798f45b6dc-2xg2d Running
html-demo-798f45b6dc-9jvlc Running
Engine running
Kubernetes running RAM 2.89 GB CPU 10.26% Disk: 5.07 GB used (limit 1006.85 GB)
Terminal
v4.66.1

Also, the HTML page should open in the port mentioned in deployment.yaml file: nodePort: 30010
So check if the HTML file open in http://localhost:30010






Divyasri M | Software Engineer
localhost:30010
Divyasri.M
About
Education
Skills
Projects
Publications
GPA: 9.20 / 10
June 2021 – May 2023
Higher Secondary
Cluny Matriculation Higher Secondary School, Vellore
Grade: 94%
Technical Skills
Languages
Java, JavaScript, C, C++, Python
Web & Mobile
HTML, CSS, React, Spring Boot, Android (Java, XML)
Databases
MySQL, Oracle SQL, Firebase Realtime DB
Machine Learning
Supervised Learning, Unsupervised Learning, CNN
Embedded / IoT
Arduino Uno, ESP8266, ESP32
Cloud & Tools
AWS, Microsoft Azure, Git, GitHub, GitHub Actions, Postman
Soft Skills
Communication, Teamwork, Problem Solving, Adaptability, Leadership, Time Management
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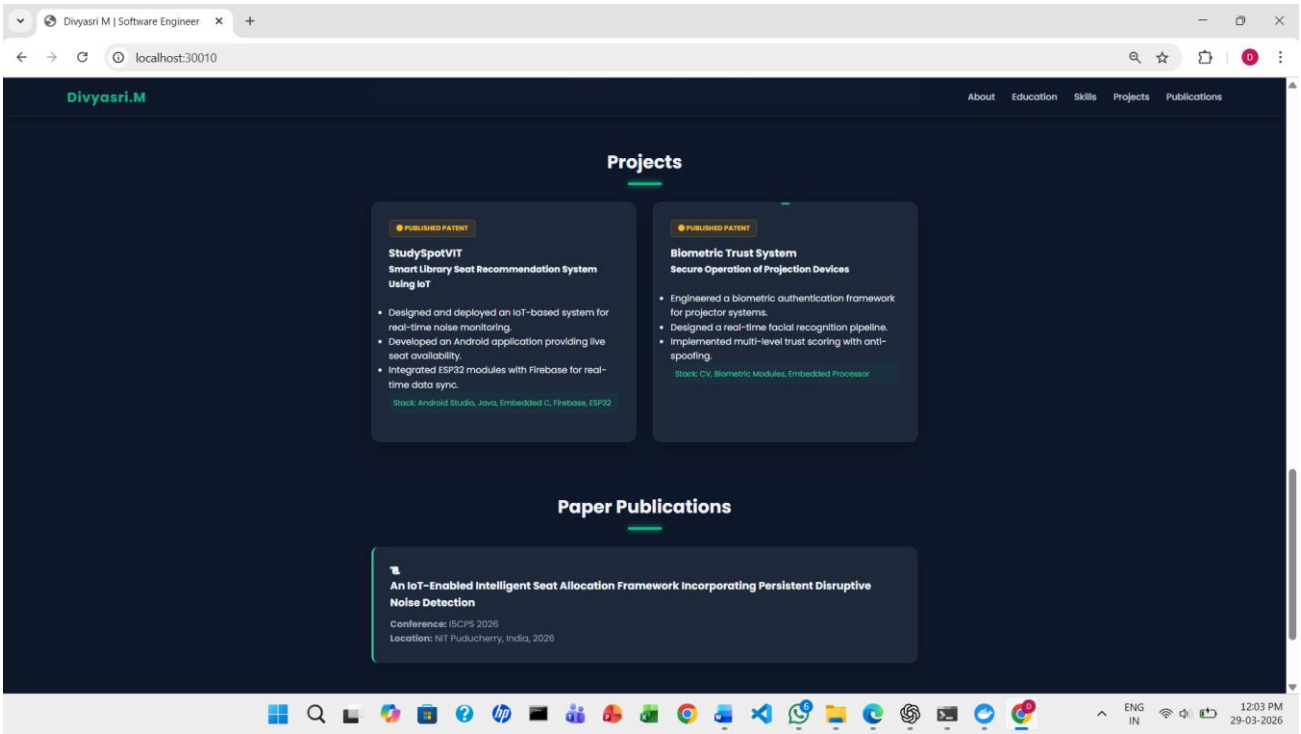
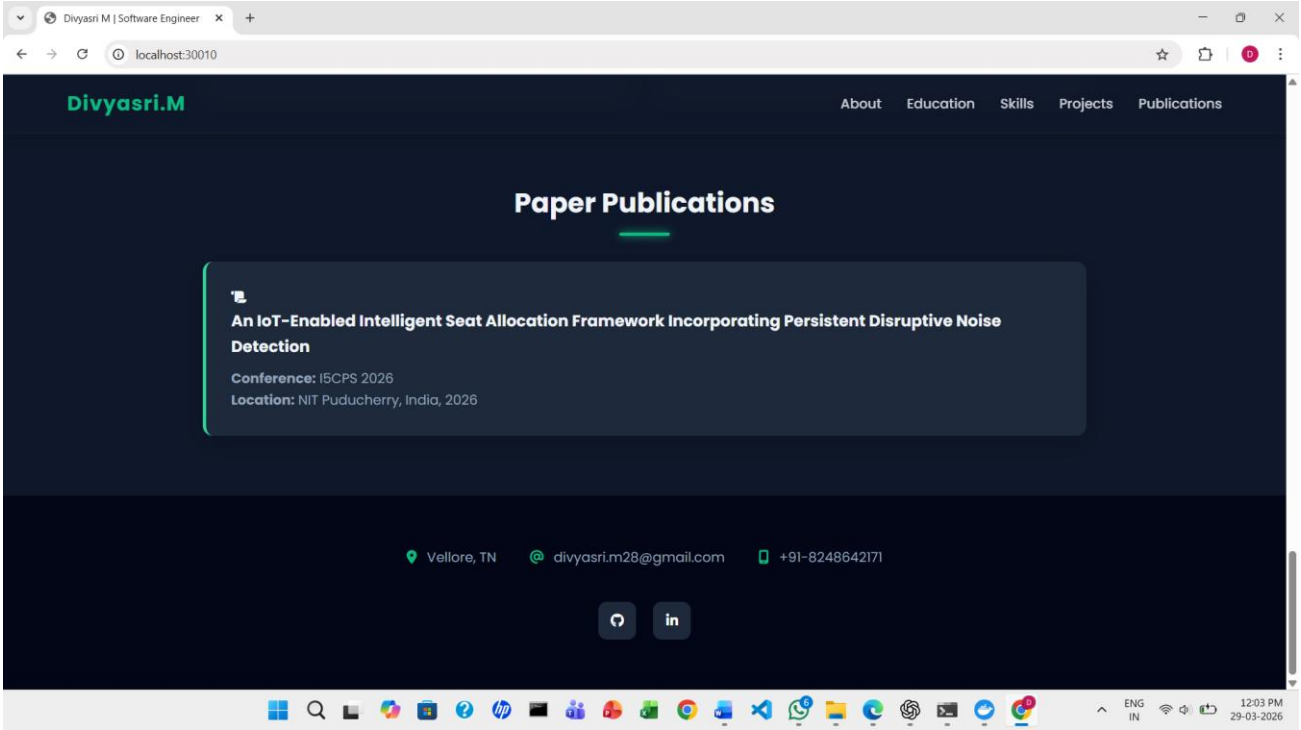

Divyasri M | Software Engineer
localhost:30010
Divyasri.M
About
Education
Skills
Projects
Publications
Projects
PUBLISHED PATENT
StudySpotVIT
Smart Library Seat Recommendation System Using IoT
Designed and deployed an IoT-based system for real-time noise monitoring.
Developed an Android application providing live seat availability.
Integrated ESP32 modules with Firebase for real-time data sync.
Stack: Android Studio, Java, Embedded C, Firebase, ESP32
PUBLISHED PATENT
Biometric Trust System
Secure Operation of Projection Devices
Engineered a biometric authentication framework for projector systems.
Designed a real-time facial recognition pipeline.
Implemented multi-level trust scoring with anti-spoofing.
Stack: CV, Biometric Modules, Embedded Processor
Paper Publications
An IoT-Enabled Intelligent Seat Allocation Framework Incorporating Persistent Disruptive Noise Detection
Conference: I5CPS 2026
Location: NIT Puducherry, India, 2026
ENG IN
12:03 PM
29-03-2026


Divyasri M | Software Engineer
localhost:30010
Divyasri.M
About
Education
Skills
Projects
Publications
Paper Publications
An IoT-Enabled Intelligent Seat Allocation Framework Incorporating Persistent Disruptive Noise Detection
Conference: I5CPS 2026
Location: NIT Puducherry, India, 2026
Vellore, TN
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ENG
IN
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